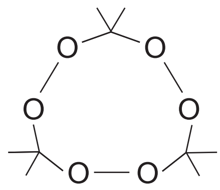
Examiner only

SECTION A

Answer **all** *questions in the spaces provided.*

1. The skeletal formula of the compound TATP is shown below.



Give the **molecular** formula of this compound. [1]

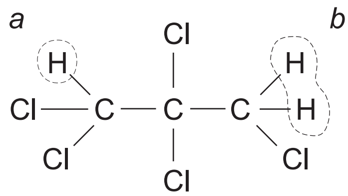
....

2. Alkanes are produced when carboxylic acids are strongly heated with sodalime.

State the name of a carboxylic acid that will produce propane, when heated in this way. [1]

....

3. Complete the table below, which describes the ^{1}H NMR spectrum of 1,1,2,2,3-pentachloropropane. [2]



Proton(s)	Splitting pattern	Relative peak area ratio
a		
b		

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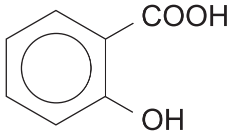
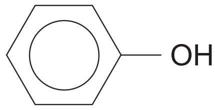
4. Using the letters provided, place these three compounds in order of **increasing** acidity. [1]



A

$(CH_3)_3N$

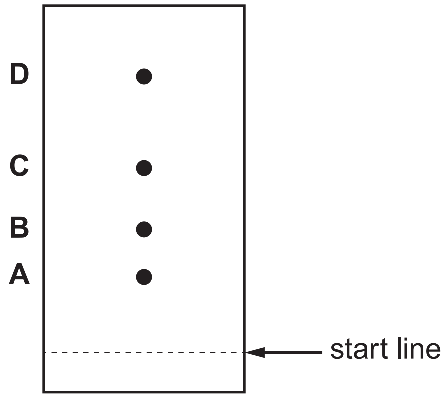
B



C

least acidic .... most acidic

5. A student produced the following chromatogram of a mixture of four amino acids.
The student forgot to mark the position of the solvent front.



The R_f values of the four amino acids are as follows.

Amino acid	R_f value
alanine	0.59
glycine	0.42
leucine	0.88
serine	0.36

State and explain which of the spots is likely to be alanine. [1]

....

....

....

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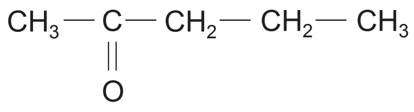
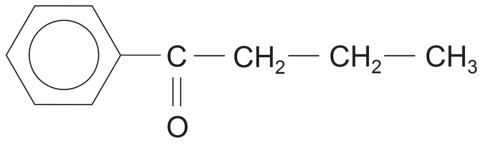
6. Describe a simple chemical test that will distinguish between pentan-2-one and 1-phenylbutanone. [2]



pentan-2-one



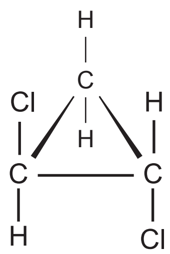
1-phenylbutanone

Reagent(s) used ....

Observations with both compounds ....

....

7. The displayed formula of 1,2-dichlorocyclopropane is given below.



(a) Draw the displayed formula of another structural isomer of formula $C_3H_4Cl_2$. [1]

(b) Draw the displayed formula of a stereoisomer of 1,2-dichlorocyclopropane. [1]

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